Incorrect forecasts for PID controllers

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*) B. Bernhardsson, K.J. Åström Control System Design – PID Control , 2007

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▲ 2002: Desborough and Miller (Honeywell):
Based on a survey of over 11000 controllers in the refining, chemicals and pulp and paper industries, 98% of regulatory controllers utilise PID feedback

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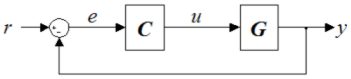
Why it is worthwhile to consider PI/PID controllers

- The vast majority of regulators in practical control systems (not only drive systems) are PI/PID controllers.
- Unlike other controllers, they can easily be set up experimentally, even if the control unit is not fully understood; the rules of procedure are (should be) well known by the engineers.
- Relatively easy: Self-adjustment and adaptation
- The results obtained in the system with PI/PID controller can be used as a basis for comparison with other control methods.

How not to think about controllers



$$C = f(G_m)$$

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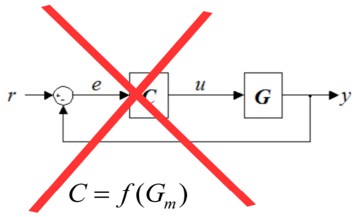

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- incomplete structure
- only reaction r → y
- Non-alternity of settings

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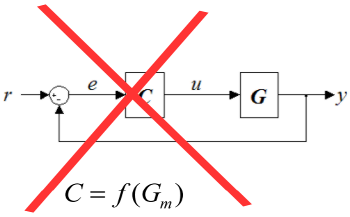

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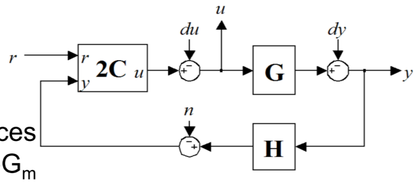

- two degrees of freedom (2DOF)
- First: $C_{y\text{->}u}$
  - reaction to disturbances
  - sensitivity to change G_m
  - compromise between speed and robustness*)
- $C_{r\text{->}u}$ less critical

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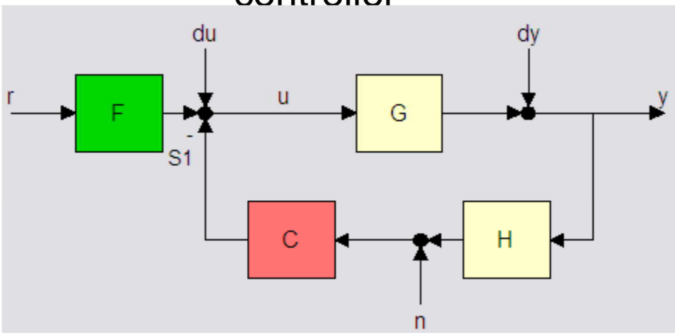
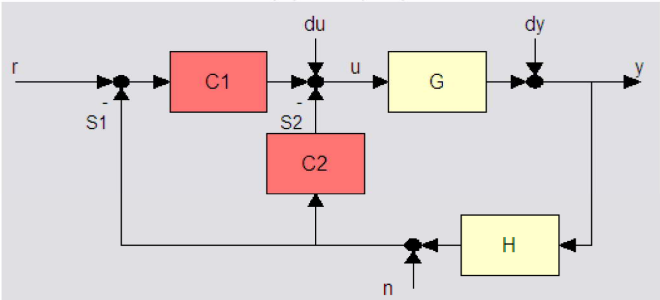
Several structures of the 2DOF controller



a

Matlab7 / Control System Toolbox / Siso Design Tool

Several structures of the 2DOF controller



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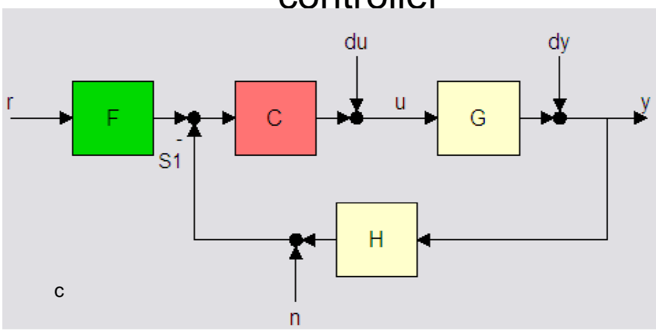
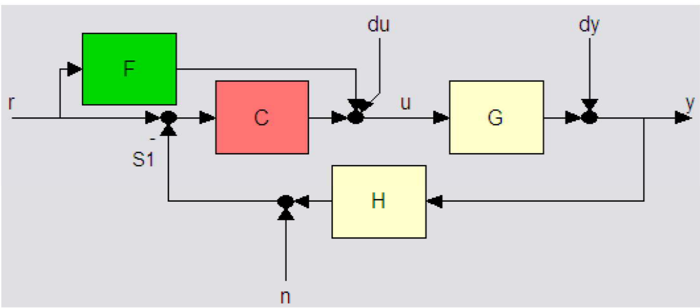


c

Matlab7 / Control System Toolbox / Siso Design Tool

Several structures of the 2DOF controller



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ISA: International Society of Automation

Standard forms of PID controller - applications in industrial controllers..

$$E(s) = Y_{sp}(s) - Y(s)$$

Form I:

$$U = K\left(bY_{sp} - Y + \frac{1}{sT_i}E + \frac{sT_d}{1 + sT_d/N}(cY_{sp} - Y)\right)$$

Weighting coefficients of the reference signal

Different definitions of error for parts P, I, D.

$$e_p = b y_{sp} - y$$

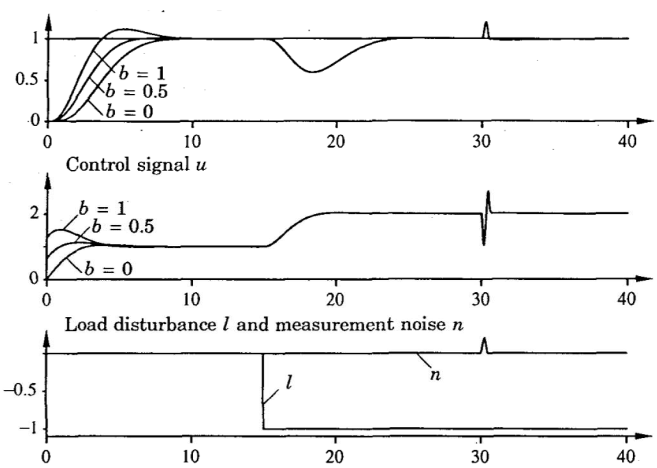
$$e_d = c y_{sp} - y$$

$$e = y_{sp} - y$$

In practice, the most common are *c = 0*

Coefficients b and c influence the system response to the set signal without changing the disturbance response

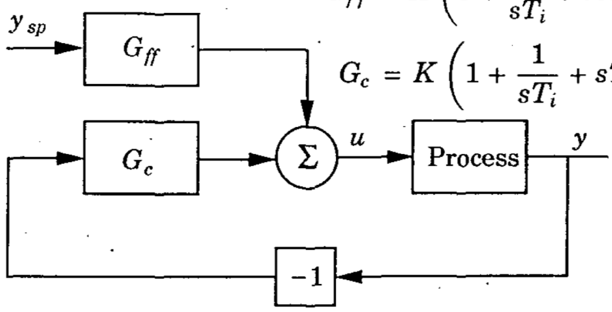
The effect of the factor b



2 DOF controller

$$G_{ff} = K\left(b + \frac{1}{sT_i} + csT_d\right)$$

$$G_c = K\left(1 + \frac{1}{sT_i} + sT_d\right)$$

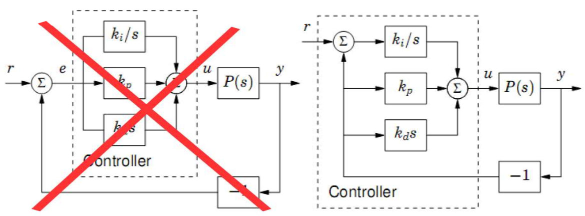


b=1 and *c*=0 – PI-D

b=0 and *c*=0 – I-PD

(‘Poor man’s 2DOF’ *)

$$U(s) = k_p(bR(s) - Y(s)) + \frac{k_i}{s}(R(s) - Y(s)) + k_d s(cR(s) - Y(s))$$



$b = c = 1$ $b = c = 0$

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Effects of measurement noise

Sinusoidal measurement noise

$$n = a \sin \omega t$$

Control signal component resulting from measurement noise

$$u_n = KT_d \frac{dn}{dt} = aKT_d\omega \cos \omega t$$

For high frequencies, the u_n component has a high amplitude. Limitation by low pass filter.

Time form of the part D equation for c=0

$$D = -\frac{T_d}{N}\frac{dD}{dt} - KT_d\frac{dy}{dt}$$

The operator equation of part D for c=0

$$D = -\frac{sKT_d}{1 + sT_d/N} y$$

Time constant of the 1st order filter: T_d/N

Maximum amplification of measurement noise: KN

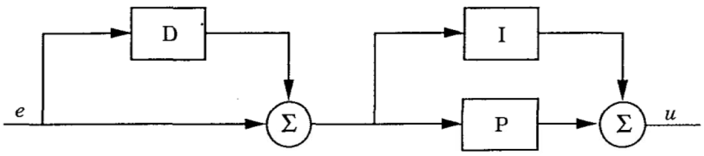
Typical values N: 8 - 20

Alternative forms of the PID controller

Form II (serial):

$$U = K'\left(\left(b + \frac{1}{sT_i'}\right)\frac{1 + scT_d'}{1 + sT_d'/N}Y_{sp} - \left(1 + \frac{1}{sT_i'}\right)\frac{1 + sT_d'}{1 + sT_d'/N}Y\right)$$

$$G'(s) = K'\left(1 + \frac{1}{sT_i'}\right)(1 + sT_d')$$



Form III (parallel):

$$U = K''(bY_{sp} - Y) + \frac{K_i''}{s}E + \frac{K_d''s}{1 + sK_d''/(NK'')}(cY_{sp} - Y)$$

$$G''(s) = k + \frac{k_i}{s} + sk_d$$

$$k = K$$

$$k_i = \frac{K}{T_i}$$

$$k_d = KT_d$$

Relationships between the parameters of the forms I and II

$$K = K'\frac{T_i' + T_d'}{T_i'} \qquad K' = \frac{K}{2}\left(1 + \sqrt{1 - 4T_d/T_i}\right)$$

$$T_i = T_i' + T_d' \qquad T_i' = \frac{T_i}{2}\left(1 + \sqrt{1 - 4T_d/T_i}\right)$$

$$T_d = \frac{T_i'T_d'}{T_i' + T_d'} \qquad T_d' = \frac{T_i}{2}\left(1 - \sqrt{1 - 4T_d/T_i}\right)$$

Parameters of industrial controllers

Controller	Structure	Setpoint weighting b	c	Derivative gain limitation N	Sampling period (s)
Allen Bradley PLC 5	I, III	1.0	1.0	none	load dependent
Bailey Net 90	II, III	0.0 or 1.0	0.0 or 1.0	10	0.25
Fisher Controls Provox	II	1.0	0.0	8	0.1, 0.25, or 1.0
Fisher Controls DPR 900, 910	II	0.0	0.0	8	0.2
Fisher Porter Micro DCI	II	1.0	0.0 or 1.0	none	0.1
Foxboro Model 761	II	1.0	0.0	10	0.25
Honeywell TDC	II	1.0	1.0	8	0.33, 0.5, or 1.0
Moore Products Type 352	II	1.0	0.0	1 – 30	0.1
Alfa Laval Automation ECA40, ECA400	II	0.0	0.0	8	0.2
Taylor Mod 30	II	0.0 or 1.0	0.0	17 or 20	0.25
Toshiba TOSDIC 200	II	1.0	1.0	3.3 – 10	0.2
Turnbull TCS 6000	II	1.0	1.0	none	0.036 – 1.56
Yokogawa SLPC	I	0.0 or 1.0	0.0 or 1.0	10	0.1

The PI regulator is sufficient

- when the dynamics of the object is well approximated by a 1-order element (level in a single tank, mixing process,...);
- for higher-order objects, when the dynamic requirements are not very strict - part I provides the elimination of a fixed error, part P provides the appropriate dynamics of transition processes.

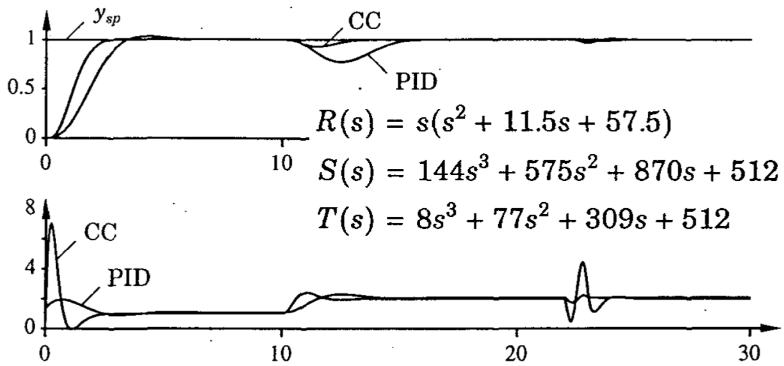
The PID regulator is sufficient

- when the dynamics of the object is well approximated by a 2-order element;
-
- when the time constants significantly differ (temperature control); segment D accelerates the reaction of the control system;
- for higher-order objects, the allowed gain of part P is limited; term D improves the damping and allows the gain to be increased.

The PID regulator is not efficient enough

- For higher-level objects

$$R(s)u(t) = -S(s)y(t) + T(s)y_{sp}(t) \qquad G(s) = \frac{1}{(s+1)^3}$$



$$R(s) = s(s^2 + 11.5s + 57.5)$$
$$S(s) = 144s^3 + 575s^2 + 870s + 512$$
$$T(s) = 8s^3 + 77s^2 + 309s + 512$$

- For objects with long dead time

$$\frac{dy(t)}{dt} = -0.5y(t) + 0.5u(t-4)$$

SP - controller with Smith predictor